Yanghao Chen

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EDUCATION

Tongji University — Shanghai, China
B.Eng. in Engineering Mechanics — Sep. 2023 – Jun. 2027 (expected)

- **GPA:** 87/100
- **Core Courses:** Theoretical Mechanics, Mathematical Analysis, Partial Differential Equations
- **Research Interests:** Computational Mechanics, Scientific Machine Learning, Differentiable Programming

RESEARCH EXPERIENCE

Johns Hopkins University — Baltimore, MD, USA
Research Intern

Hybrid Finite Element–Neural Operator Solver for Solid Mechanics — Jan. 2026 – Present
Advisor: Prof. Somdatta Goswami

- Developing a hybrid finite element–neural operator solver for solid mechanics using non-overlapping domain decomposition. FEM solves the outer subdomain, while a data-driven neural operator predicts stress fields in the inner subdomain. The two solvers exchange interface displacements and reaction forces iteratively.
- Generated training data by sampling interface displacements with Gaussian random fields and computing corresponding subdomain stresses with GPU-accelerated JAX-FEM. Trained Transolver in JAX to learn the mapping from interface displacements to subdomain stress fields.
- Investigated convergence acceleration and interface relaxation strategies for the coupled solver. Extending the framework to 3D fiber-reinforced composites, using FEM for the matrix subdomain and neural operators with model weights shared across all fiber subdomains.

Tongji University — Shanghai, China
Undergraduate Researcher

Physics-Informed Neural Networks for Shock Capturing — Nov. 2025 – Mar. 2026
Advisor: Prof. Xianyang (Tom) Chen

- Trained a physics-informed neural network (PINN) in PyTorch for the one-dimensional inviscid Burgers shock problem, using PDE residual, initial condition, and boundary condition losses.
- Trained a second PINN by adding an artificial-viscosity term with a fixed coefficient to the PDE residual, then compared both formulations in predicting the sharp shock transition.
- Evaluated both PINNs against the analytical inviscid solution using shock location, transition width, relative L^2 error, and maximum pointwise error to assess shock resolution and prediction accuracy.

ConvLSTM Modeling of Chemo-Mechanical Fields in Battery Materials — Jun. 2025 – Oct. 2025
Advisor: Prof. Ying Zhao

- Generated polycrystalline NMC microstructures and automated MATLAB–COMSOL simulations to construct time-aligned image sequences of lithium concentration and von Mises stress for model training and evaluation.
- Trained separate conditional ConvLSTM models in PyTorch for concentration and stress prediction using past field images, grain-orientation maps, and C-rate inputs, with MSE–SSIM losses and scheduled sampling.
- Evaluated autoregressive concentration and stress predictions against COMSOL reference images to assess spatial accuracy and error accumulation over successive forecast steps.

Instruction Fine-Tuning for Mechanics of Materials — Sep. 2024 – Apr. 2025
Advisor: Prof. Ying Zhao

- Constructed instruction datasets from textbooks and technical literature in mechanics of materials, covering stress analysis, constitutive relations, and failure theories.
- Fine-tuned Qwen2.5-7B using low-rank adaptation (LoRA) on the constructed instruction and response pairs for question answering and concept explanation in mechanics.
- Released two instruction datasets and two Qwen2.5 model versions on Hugging Face, including the fine-tuned checkpoint and a merged model for standalone inference.

TECHNICAL SKILLS

Scientific Machine Learning: PyTorch, JAX, Neural Operators, PINNs, ConvLSTM, Fine-tuning (LoRA)
Numerical Simulation: JAX-FEM, FEniCSx, Abaqus, COMSOL Multiphysics
Programming & Tools: Python, C/C++, MATLAB, Linux, Git, Matplotlib, Origin
Languages: Mandarin Chinese (native), Cantonese, English